

1(a)

feature	animal cell	plant cell
80S ribosomes in the cytoplasm	✓	✓
cell wall composed of peptidoglycan	✗	✗
large permanent vacuole	✗	✓

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one mark per correct row
if no marks gained, allow 1 mark for a correct column

3

Question	Answer	Marks
1(b)(i)	<i>marking points must be in context of companion cell</i> any five from: 1 moves / pumps / actively transports, hydrogen ions / H⁺ / protons, into the apoplast / into the cell wall / out of cytoplasm / out of companion cell R into sieve tube or proton pump creates proton gradient between, apoplast / cell wall, and cytoplasm / AW ; 2 cotransporter protein for cotransport of sucrose and, hydrogen ions / H⁺ / protons, into, cell A / companion cell ; 3 plasmodesmata allow, diffusion / transfer / AW, of sucrose (from, cell A / companion cell) into (phloem) sieve tube (element) ; 4 mitochondria produce ATP ; 5 (many) ribosomes / rough endoplasmic reticulum, to produce (named) transport proteins / AW ; 6 nucleus contains, genes / AW, coding for transport proteins ; 7 AVP ; e.g. example of need for ATP including for, proton pumping for synthesis of, transport proteins for synthesis of, phloem / P, proteins	5

Question	Answer	Marks
1(b)(ii)	any four from: 1 mass flow ; 2 (entry of sucrose) decreases water potential in sieve tube ; A Ψ for water potential 3 water enters (sieve tubes), down water potential gradient / by osmosis ; 4 increase in volume ; 5 increase in / high(er), hydrostatic pressure ; <i>hydrostatic not needed if given in mp 6</i> 6 unloading / removal, of sucrose and water at sink lowers hydrostatic pressure ; <i>hydrostatic not needed if given in mp5</i> 7 movement (from source to sink) is down (hydrostatic) pressure gradient ;	4